EC442 Macroeconomics for MRes students | Instructor: Matthias Doepke
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Problem Set 3

Dynamic Programming: Practice

Question 1: In this question, you will use polynomial approximation to numerically solve the planning problem from Question 1 of Problem Set 2. Consider the Bellman equation for the social planning problem for the "standard model:"

$$v(k) = \max_{k'} \left\{ \log\left((1-\delta)k + Ak^{\alpha} - k'\right) + \beta v(k') \right\}.$$

We will once again use the parameter values $\beta = 0.9$, $A = 3$, $\alpha = 0.3$, and $\delta = 0.2$.

- We will approximate the value function by polynomials of the form:

$$v(k) = \sum_{i=0}^{n} a_i k^i.$$

  The objective is to find coefficients a_i that yield a good approximation to the value function defined by the Bellman equation for a given order n of the polynomial.

- Start by fixing n and making an initial guess for the coefficients $a_{i,0} = 0$ for all i, so that the initial guess for the value function is:

$$v_0(k) = \sum_{i=0}^{n} a_{i,0} k^i = 0.$$

- Then use the Bellman equation (just as in discrete-state dynamic programming) to compute a new guess $v_1(k)$ for the value function, i.e., a new set of coefficients $a_{i,1}$. What is convenient about using a parametric function to approximate the value function is that we need to evaluate the Bellman equation only at a few points to make a new guess. Specifically, we need to know $v_1(k)$ at $n+1$ distinct points (which we will refer to as "nodes") to be able to solve for the coefficients $a_{i,1}$. We therefore first need to fix a set of nodes K. As a first pass, use at least $n+1$ evenly spaced points in the interval $[1, 10]$ as your set of nodes (this choice can in principle be optimized). For each capital value $k \in K$ at a node, compute a new guess for the value function by solving:

$$v_1(k) = \max_{k'} \left\{ \log((1-\delta)k + Ak^{\alpha} - k') + \beta v_0(k') \right\}.$$

It is up to you to decide how to carry out the maximization step. You could do something simple such as a grid search, or use an existing routine such as `fminbnd` or `fmincon` in MATLAB or something similar in your language of choice.

- The maximization step results in one value for the new guess of the value function $v_1(k)$ at each of the nodes $k \in K$. Use these values to find new polynomial coefficients $a_{i,1}$ that fit these points. Since the coefficients enter linearly, with exactly $n+1$ nodes the polynomial interpolates the points and the coefficients solve a system of $n+1$ linear equations; with more nodes, they solve a linear least-squares problem. The backslash operator in MATLAB handles both cases, as does `numpy.linalg.lstsq` in Python.

- Continue constructing new guesses of the value function in this way, until the convergence criterion:
$$\max_i |a_{i,j+1} - a_{i,j}| < 10^{-5}$$
is met, where j indexes the iterations.

Compute polynomial approximations to the value function for $n = 2, 3, \ldots, 6$:

1. Plot the resulting approximations on the interval $k \in [0, 15]$ together with the computed value function using the Howard algorithm from Problem Set 2.
2. Provide a second plot of the approximated policy functions, once again together with the policy function resulting from the Howard improvement algorithm.
3. How large does the order of the polynomial have to be to yield a reasonable approximation?

A Model with Human Capital

Question 2: Consider an economy in which the representative consumer has preferences:
$$\sum_{t=0}^{\infty} \beta^t \frac{c_t^{1-\sigma}}{1-\sigma},$$
where $\sigma > 0$ and $\sigma \neq 1$.

The production function is given by:
$$y_t = A k_t^{\alpha} h_t^{1-\alpha}.$$

The laws of motion for physical and human capital are:

$$k_{t+1} = (1-\delta)k_t + x_{kt},$$
$$h_{t+1} = (1-\delta)h_t + x_{ht}.$$

Output can be converted one-to-one into consumption, physical investment, or human-capital investment, so that the resource constraint is

$$c_t + x_{kt} + x_{ht} = y_t.$$

Assume that $\beta\left(A\alpha^{\alpha}(1-\alpha)^{1-\alpha} + 1 - \delta\right)^{1-\sigma} < 1$.

1. Formulate the Bellman equation for the social planning problem, and show that the value function is homogeneous (of which degree?).
2. Compute the balanced growth rate of the economy, and characterize the transitional dynamics for arbitrary initial levels of k_0 and h_0.

An AK Model with Endogenous Labor Supply

Question 3: Consider an economy in which the representative consumer has preferences:

$$\sum_{t=0}^{\infty} \beta^t (c_t + \log(1-n_t)),$$

where c_t is consumption, n_t is labor supply, and $1-n_t$ is leisure. Capital k_t depreciates completely every period. There are separate technologies for producing capital and the consumption good; in particular, capital is produced using capital and labor, whereas consumption is produced using only capital:

$$c_t = Ak_t,$$
$$k_{t+1} = k_t n_t.$$

Notice that the same capital is used in both sectors; the consumer only has to decide on labor supply, but not on the allocation of capital. The consumer also has to obey the constraint $0 \leq n_t < 1$, and the initial capital stock $k_0 > 0$ is given.

1. Formulate the Bellman equation for the social planning problem.
2. Specify an appropriate state space, and show that the Bellman equation satisfies Blackwell's sufficient conditions for a contraction.
3. Prove that the value function is increasing and convex, and explain the economic intuition underlying your result.
4. Characterize the optimal decision rule as sharply as possible. Show that the capital stock reaches zero in finite time.